

From: Schmitt, Glen (EGLE)
Sent: 11/22/2019 9:18:01 AM
To: "Brady, Donal (EGLE)" <BradyD6@michigan.gov>; "Alexander, Christine (EGLE)" <ALEXANDERC2@michigan.gov>
Cc: "Lowe, Claire (EGLE)" <LoweC9@michigan.gov>
Subject: RE: Final Report for City of Cadillac WWTP - Outfall 001A

Hi Don!

WET Testing for Fathead minnow was included in the previous permit as well as part of the Additional Monitoring Requirements. With WET tests, the test species are representative of the fish community and the macroinvertebrate community. All POTW's that have Additional Monitoring Requirements test with both Fathead minnow and either *C. dubia* or *D. magna*.

For Cadillac's issues, it is hard to say what is causing the toxicity based on the information provided. Was any monitoring of the VOC/SVOC's completed of the industrial users? Sometimes it may not be a single constituent but could be a combination of things that cause the toxicity. I would suggest they follow-up with additional tests for both species in December since now we have toxicity on both test organisms. While I understand the October test was not toxic to *C. dubia*, additional testing is still warranted.

Let me know if you should have any questions. Thanks!

Glen P. Schmitt
Aquatic Biology Specialist
Water Quality and Aquatic Nuisance Control Permits Unit
Permits Section, Water Resources Division
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From: Brady, Donal (EGLE) <BradyD6@michigan.gov>
Sent: Thursday, November 21, 2019 5:01 PM
To: Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>
Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Subject: FW: Final Report for City of Cadillac WWTP - Outfall 001A

Hi Glen and Chris,

The August 2019 WET tests for the Cadillac WWTP exceeded 1.0 TU for/with water fleas. Follow-up TIE investigation indicated the toxicity may have been related to VOCs or SVOCs. An October 2019 WET test with water fleas indicated no toxicity (acute or chronic) but did indicate chronic toxicity with fathead minnow (1.1 TU). WET testing with fathead minnows is a new permit requirement and was therefore not completed in August. They are planning to do some additional TIE testing related to the August sample and conduct another WET test in early December 2019 for chronic minnow toxicity.

Can you please tell me or point me to info. explaining why WET testing with fathead minnows was added to the NPDES permits? Was it something specific or just generally mo species mo better? Do you know of any common chemicals that are generally more toxic to minnows than water fleas (I can't find much on-line in this regard)? Thanks!

Kind Regards,



MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY

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From: Brady, Donal (EGLE)
Sent: Friday, September 13, 2019 1:42 PM
To: Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>; Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Popovich, Bradley (EGLE) <PopovichB@michigan.gov>
Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>; Dijak, Claire (EGLE) <DijakC@michigan.gov>; Heaton, Sylvia (EGLE) <HEATONS@michigan.gov>; Lane, Byron (EGLE) <LANEB@michigan.gov>
Subject: RE: Final Report for City of Cadillac WWTP - Outfall 001A

Hi Chris and Glen,

Thanks for the input.

O.K. on the monthly sampling, I will push for that at the plant.

I will talk to IPP staff (Claire L.) about this. I think surveying industrial users is a good idea and will also push for that.

I understand a recent WWTP upset in Claire D.'s district (with some common characteristics with Cadillac) was related to zinc. The Cadillac WWTP has also experienced a past upset related to zinc. Depending on what Claire L. thinks, we may also push for additional monitoring of zinc (and/or other suspects identified in the Phase I TIE).



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From: Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>
Sent: Friday, September 13, 2019 12:51 PM
To: Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>; Brady, Donal (EGLE) <BradyD6@michigan.gov>; Popovich, Bradley (EGLE) <PopovichB@michigan.gov>
Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>; Dijak, Claire (EGLE) <DijakC@michigan.gov>; Heaton, Sylvia (EGLE) <HEATONS@michigan.gov>; Lane, Byron (EGLE) <LANEB@michigan.gov>
Subject: RE: Final Report for City of Cadillac WWTP - Outfall 001A

Hi everyone.

Thanks for keeping us in the loop.

Don—are the IPP folks involved by chance? I'm just wondering if anyone has checked with the indirect user on process changes, tank clean outs, etc.? We've seen toxicity caused by changes in cleaners, or random use of other products that have caused toxicity in the effluent of the WWTP. Makes me wonder about the intermittent upsets described below.

Just my two cents.

Thanks-
Chris

Christine Alexander, Manager
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From: Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>
Sent: Friday, September 13, 2019 9:58 AM
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Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>; Dijak, Claire (EGLE) <DijakC@michigan.gov>; Heaton, Sylvia (EGLE) <HEATONS@michigan.gov>; Lane, Byron (EGLE) <LANEB@michigan.gov>; Alexander, Christine (EGLE) <ALEXANDERC2@michigan.gov>
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Hi Donal!

For the time being, I would recommend monthly monitoring for WET until this issue is resolved. The two chronic results of 1.6 TUC and 5.6 TUC exceed the allowable level if a limit for Chronic WET was included in the permit. The previous permit had a Chronic WET limit of 1.1 TUC. The limit went away because they demonstrated compliance with 1.1 TUC.

I guess it is bad timing the permit was just issued and the facility all of sudden has effluent that is chronically toxic. However, it is encouraging to see they are conducting the Phase I TIE to try and identify what is causing the toxicity. I am not sure if Byron and Chris want to weigh in on what other options there might be.

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From: Brady, Donal (EGLE) <BradyD6@michigan.gov>
Sent: Tuesday, September 10, 2019 1:24 PM
To: Popovich, Bradley (EGLE) <PopovichB@michigan.gov>; Schmitt, Glen (EGLE) <SCHMITTG1@michigan.gov>
Cc: Lowe, Claire (EGLE) <LoweC9@michigan.gov>; Dijak, Claire (EGLE) <DijakC@michigan.gov>
Subject: FW: Final Report for City of Cadillac WWTP - Outfall 001A

Hi Bradley and Glen,

Thanks for your input on this.

The Cadillac WWTP conducted quarterly WET testing the week of August 13 (in accordance with the former permit). The results are attached. No acute toxicity was observed but chronic toxicity of 1.6 units was identified. The plant observed more filamentous bacteria and a slight reduction in settleability at this same time. The plant operators responded to this by increasing the sludge and scum removal from the primary clarifiers.

The plant collected additional WET test samples the week of August 27. These results are pending. Preliminary test results of the resample indicated no acute toxicity but chronic toxicity of 5.6 units. During the week of August 27 the WWTP collected enough sample for additional testing. The lab (ERM) is going to conduct Phase I of a Toxicity Identification Evaluation (please see the attached). The City is going to let me know when the results of the additional testing will be available and also will provide more details on the Toxicity Identification Evaluation to be performed. Interestingly, the plant bacteria and settleability appeared to be back to normal the week of August 27.

The operators remembered two past upsets. During one event (approximately 5 years ago) the plant's population of nitrifying bacteria was killed (and a replacement population was also killed). The plant experienced high ammonia in the effluent during this event, this event was short lived, and the cause never identified. During another, older event the upset was traced to zinc from an industrial source.

Approximately 40% of this plant's flow is industrial. I understand the plant monitors its influent for CBOD, TSS, ammonia, phosphorus, cyanide, copper, mercury, and pH and that no anomalies with the August influent were identified.

Please let me and the Claires know if you have any particular questions, thoughts, and/or recommendations at this time. I will send the additional testing information upon receipt. Thanks!



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From: Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Sent: Tuesday, September 10, 2019 11:04 AM
To: Brady, Donal (EGLE) <BradyD6@michigan.gov>
Cc: Doug Langworthy <dlangworthy@Cadillac-MI.net>; Jeff Dietlin <utilities@cadillac-mi.net>
Subject: FW: Final Report for City of Cadillac WWTP - Outfall 001A

Donal –

This is the final report from the original set of August WET tests.

I will send the final report for the second set of August WET tests as soon as we receive it. I will also forward the additional info regarding the Toxicity Investigation Evaluation (TIE) as soon as I hear back from our lab.

Thanks,

Cindy A. Tomaszewski
Lab Supervisor

From: Kaila Maitland <Kaila.Maitland@erm.com>
Sent: Wednesday, September 4, 2019 9:55 AM
To: Doug Langworthy <dlangworthy@Cadillac-MI.net>
Cc: Bruce Rabe <Bruce.Rabe@erm.com>; Cindy Tomaszewski <ctomaszewski@Cadillac-MI.net>
Subject: Final Report for City of Cadillac WWTP - Outfall 001A

Good morning,

Attached please find the final report containing the results of the August 2019 WET test performed on samples of Outfall 001A effluent. Please let me know if you have any questions regarding this report.

Have a great day!

Kaila Maitland

Administrative Assistant/Staff Scientist

ERM

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